Q1- In the reactions given below, identify the species undergoing oxidation and reduction:
(i) H_2S (g) + Cl_2 (g) → 2 HCl (g) + S (s)
(ii) $3Fe_3O_4$ (s) + 8 Al (s) → 9 Fe (s) + $4Al_2O_3$ (s)
(iii) 2 Na (s) + H_2 (g) → 2 NaH (s)

Q2- Justify that the reaction: 2 Na(s) + H_2(g) → 2 NaH (s) is a redox change.

Q3- Using Stock notation, represent the following compounds :$HAuCl_4$, Tl_2O, FeO, Fe_2O_3, CuI, CuO, MnO and MnO_2.

Q4- Justify that the reaction: $2Cu_2O$(s) + Cu_2S(s) → 6Cu(s) + SO_2(g) is a redox reaction. Identify the species oxidised/reduced, which acts as an oxidant and which acts as a reductant

Q5- Which of the following species, do not show disproportionation reaction and why ? ClO^- , ClO_2^- , ClO_3^- and ClO_4^- Also write reaction for each of the species that disproportionates.

Q6- Why do the following reactions proceed differently ?
Pb_3O_4 + 8HCl → $3PbCl_2$ + Cl_2 + $4H_2O$ and Pb_3O_4 + $4HNO_3$ → $2Pb(NO_3)_2$ + PbO_2 + $2H_2O$

Q7- Write the net ionic equation for the reaction of potassium dichromate(VI), $K_2Cr_2O_7$ with sodium sulphite, Na_2SO_3, in an acid solution to give chromium(III) ion and the sulphate ion.

Q8- Permanganate ion reacts with bromide ion in basic medium to give manganese dioxide and bromate ion. Write the balanced ionic equation for the reaction.

Q9- Permanganate(VII) ion, MnO_4^- in basic solution oxidises iodide ion, I– to produce molecular iodine (I_2) and manganese (IV) oxide (MnO_2). Write a balanced ionic equation to represent this redox reaction.

Q10- Assign oxidation number to the underlined elements in each of the following species:
(a) NaH_2PO_4 (b) $NaHSO_4$ (c) $H_4P_2O_7$ (d) K_2MnO_4
(e) CaO_2 (f) $NaBH_4$ (g) $H_2S_2O_7$ (h) $KAl(SO_4)_2.12\ H_2O$

Q11- What are the oxidation number of the underlined elements in each of the following and how do you rationalise your results ?
(a) KI_3 (b) $H_2S_4O_6$ (c) Fe_3O_4 (d) CH_3CH_2OH (e) CH_3COOH

Q12- Justify that the following reactions are redox reactions:

(a) $CuO(s) + H_2(g) \rightarrow Cu(s) + H_2O(g)$

(b) $Fe_2O_3(s) + 3CO(g) \rightarrow 2Fe(s) + 3CO_2(g)$

(c) $4BCl_3(g) + 3LiAlH_4(s) \rightarrow 2B_2H_6(g) + 3LiCl(s) + 3\ AlCl_3(s)$

(d) $2K(s) + F_2(g) \rightarrow 2K^+F^-(s)$

(e) $4\ NH_3(g) + 5\ O_2(g) \rightarrow 4NO(g) + 6H_2O(g)$

Q13- Fluorine reacts with ice and results in the change: H_2O(s) + F_2(g) → HF(g) + HOF(g) Justify that this reaction is a redox reaction.

Q14- Calculate the oxidation number of sulphur, chromium and nitrogen in H_2SO_5, Cr_2 and NO_3^- . Suggest structure of these compounds. Count for the fallacy.

Q15- Write formulas for the following compounds:
(a) Mercury(II) chloride
(b) Nickel(II) sulphate

(c) Tin(IV) oxide
(d) Thallium(I) sulphate
(e) Iron(III) sulphate
(f) Chromium(III) oxide

Q16- Suggest a list of the substances where carbon can exhibit oxidation states from –4 to +4 and nitrogen from –3 to +5.

Q17- While sulphur dioxide and hydrogen peroxide can act as oxidising as well as reducing agents in their reactions, ozone and nitric acid act only as oxidants. Why ?

Q18- Consider the reactions:
(a) $6\ CO_2(g) + 6H_2O(l) \rightarrow C_6H_{12}O_6(aq) + 6O_2(g)$
(b) $O_3(g) + H_2O_2(l) \rightarrow H_2O(l) + 2O_2(g)$
Why it is more appropriate to write these reactions as :
(a) $6CO_2(g) + 12H_2O(l) \rightarrow C_6H_{12}O_6(aq) + 6H_2O(l) + 6O_2(g)$
(b) $O_3(g) + H_2O_2\ (l) \rightarrow H_2O(l) + O_2(g) + O_2(g)$
Also suggest a technique to investigate the path of the above (a) and (b) redox reactions.

Q19- The compound AgF2 is unstable compound. However, if formed, the compound acts as a very strong oxidising agent. Why ?

Q20- Whenever a reaction between an oxidising agent and a reducing agent is carried out, a compound of lower oxidation state is formed if the reducing agent is in excess and a compound of higher oxidation state is formed if the oxidising agent is in excess. Justify this statement giving three illustrations.

Q21- How do you count for the following observations ?
(a) Though alkaline potassium permanganate and acidic potassium permanganate both are used as oxidants, yet in the manufacture of benzoic acid from toluene we use alcoholic potassium permanganate as an oxidant. Why ? Write a balanced redox equation for the reaction.
(b) When concentrated sulphuric acid is added to an inorganic mixture containing chloride, we get colourless pungent smelling gas HCl, but if the mixture contains bromide then we get red vapour of bromine. Why ?

Q22- Identify the substance oxidised reduced, oxidising agent and reducing agent for each of the following reactions:

(a) $2AgBr\ (s) + C_6H_6O_2(aq) \quad 2Ag(s) + 2HBr\ (aq) + C_6H_4O_2(aq)$
(b) $HCHO(l) + 2[Ag\ (NH_3)_2]^+(aq) + 3OH^-(aq) \quad 2Ag(s) + HCOO^-(aq) + 4NH_3(aq) + 2H2O(l)$
(c) $HCHO\ (l) + 2\ Cu^{2+}(aq) + 5\ OH^-(aq) \quad Cu2O(s) + HCOO^-(aq) + 3H_2O(l)$
(d) $N_2H_4(l) + 2H_2O_2(l) \rightarrow N_2(g) + 4H_2O(l)$
(e) $Pb(s) + PbO_2(s) + 2H_2SO_4(aq) \rightarrow 2PbSO_4(s) + 2H_2O(l)$

Q23- Consider the reactions :

$2\ S_2O_3^{2-}(aq) + I_2(s) \rightarrow S_4O_6^{2-}(aq) + 2I^-(aq)$

$S_2O_3^{2-}(aq) + 2Br_2(l) + 5\ H_2O(l) \rightarrow 2SO_4^{2-}(aq) + 4Br^-(aq) + 10H^+(aq)$

Why does the same reductant, thiosulphate react differently with iodine and bromine ?

Q24- Justify giving reactions that among halogens, fluorine is the best oxidant and among hydrohalic compounds, hydroiodic acid is the best reductant.

Q25- Why does the following reaction occur ?

$XeO_6^{4-}(aq) + 2F^-(aq) + 6H^+(aq) \rightarrow XeO_3(g) + F_2(g) + 3H_2O(l)$

What conclusion about the compound Na_4XeO_6 (of which XeO_6^{4-} is a part) can be drawn from the reaction.

(a) $H_3PO_2(aq) + 4\ AgNO_3(aq) + 2\ H_2O(l) \rightarrow H_3PO_4(aq) + 4Ag(s) + 4HNO_3(aq)$

(b) $H_3PO_2(aq) + 2CuSO_4(aq) + 2\ H_2O(l) \rightarrow H_3PO_4(aq) + 2Cu(s) + H_2SO_4(aq)$

(c) $C_6H_5CHO(l) + 2[Ag(NH_3)_2]^+(aq) + 3OH^-(aq) \rightarrow C_6H_5COO^-(aq) + 2Ag(s) + 4NH_3(aq) + 2\ H_2O(l)$

(d) $C_6H_5CHO(l) + 2Cu^{2+}(aq) + 5OH^-(aq) \rightarrow$ No change observed.

What inference do you draw about the behaviour of Ag^+ and Cu^{2+} from these reactions ?

Q26- Balance the following redox reactions by ion – electron method :

(a) $MnO_4^-(aq) + I^-(aq) \rightarrow MnO_2(s) + I_2(s)$ (in basic medium)

(b) $MnO_4^-(aq) + SO_2(g) \rightarrow Mn^{2+}(aq) + HSO_4^-(aq)$ (in acidic solution)

(c) $H_2O_2(aq) + Fe^{2+}(aq) \rightarrow Fe^{3+}(aq) + H_2O(l)$ (in acidic solution)

(d) $Cr_2O_7^{2-} + SO_2(g) \rightarrow Cr^{3+}(aq) + SO_4^{2-}(aq)$ (in acidic solution)

Q27- Balance the following equations in basic medium by ion-electron method and oxidation number methods and identify the oxidising agent and the reducing agent.

(a) $P_4(s) + OH^-(aq) \rightarrow PH_3(g) + HPO_2^-(aq)$

(b) $N_2H_4(l) + ClO_3^-(aq) \rightarrow NO(g) + Cl^-(g)$

(c) $Cl_2O_7(g) + H_2O_2(aq) \rightarrow ClO_2^-(aq) + O_2(g) + H^+$

Q28- What sorts of informations can you draw from the following reaction ? (CN)2(g) + 2OH– (aq) → CN– (aq) + CNO– (aq) + H2O(l) 7.21 The Mn3+ ion is unstable in solution and undergoes disproportionation to give Mn2+, MnO2, and H+ ion. Write a balanced ionic equation for the reaction.

Q29- Consider the elements : Cs, Ne, I and F

(a) Identify the element that exhibits only negative oxidation state.

(b) Identify the element that exhibits only postive oxidation state.

(c) Identify the element that exhibits both positive and negative oxidation states.

(d) Identify the element which exhibits neither the negative nor does the positive oxidation state.

Q30- Chlorine is used to purify drinking water. Excess of chlorine is harmful. The excess of chlorine is removed by treating with sulphur dioxide. Present a balanced equation for this redox change taking place in water.

Q31- Refer to the periodic table given in your book and now answer the following questions:
(a) Select the possible non metals that can show disproportionation reaction.
(b) Select three metals that can show disproportionation reaction.

Q32- In Ostwald's process for the manufacture of nitric acid, the first step involves the oxidation of ammonia gas by oxygen gas to give nitric oxide gas and steam. What is the maximum weight of nitric oxide that can be obtained starting only with 10.00 g. of ammonia and 20.00 g of oxygen ?

Q33- Using the standard electrode potentials predict if the reaction between the following is feasible:
(a) Fe^{3+}(aq) and I^- (aq)
(b) Ag^+(aq) and Cu(s)
(c) Fe^{3+} (aq) and Cu(s)
(d) Ag(s) and Fe^{3+}(aq)
(e) Br_2(aq) and Fe^{2+}(aq)

Q34- Predict the products of electrolysis in each of the following:
(i) An aqueous solution of $AgNO_3$ with silver electrodes
(ii) An aqueous solution $AgNO_3$ with platinum electrodes
(iii) A dilute solution of H_2SO_4 with platinum electrodes
(iv) An aqueous solution of $CuCl_2$ with platinum electrodes.

Q35- Arrange the following metals in the order in which they displace each other from the solution of their salts. Al, Cu, Fe, Mg and Zn.

Q36- Given the standard electrode potentials, K+/K = –2.93V, Ag^+/Ag = 0.80V, Hg^{2+}/Hg = 0.79V Mg2+/Mg = –2.37V. Cr^{3+}/Cr = –0.74V arrange these metals in their increasing order of reducing power.

Q37- Depict the galvanic cell in which the reaction Zn(s) + $2Ag^+$(aq) → Zn^{2+}(aq) +2Ag(s) takes place, Further show:
(i) which of the electrode is negatively charged,
(ii) the carriers of the current in the cell, and
(iii) individual reaction at each electrode